Types of RNG

True random number generation (TRNG):

- captures unpredictable physical processes to generate randomness.
- These physical processes act as an entropy source. Because the underlying physics (like quantum fluctuations or thermal noise) is inherently chaotic and non-deterministic, the outputs of a TRNG cannot be predicted or reproduced, even if an attacker has complete knowledge of the hardware design.

Types of TRNG:

- **Quantum Random Number Generation (QRNG):** Generates true randomness by exploiting the fundamentally unpredictable, non-deterministic behaviors of quantum mechanics, such as photon paths or vacuum fluctuations.
- **Thermal and Electronic Noise TRNGs:** Capitalizes on the chaotic, temperature-dependent microscopic agitation of electrons inside resistors and semiconductors to create unpredictable voltage fluctuations.
- **Oscillator-Based TRNGs:** Extracts randomness directly on silicon chips by measuring the micro-timing instability and phase jitter created as digital logic gates toggle state.
- **Radioactive Decay TRNGs:** Relies on the purely spontaneous, unpredictable timing of atomic nucleus degradation and particle emission from a radioactive isotope.
- **Atmospheric and Environmental TRNGs:** Captures macroscopic, large-scale chaotic phenomena from the physical environment, such as lightning-induced radio static or the fluid dynamics of moving wax inside lava lamps.

Pseudo random generation (PRNG):

- is an algorithm that uses mathematical formulas to produce a sequence of numbers that appears random, but is completely deterministic.
- starts with an initial input value called a seed. As long as the seed and the algorithm are known, the exact same sequence of numbers can be reproduced every single time. Because they are driven by deterministic math, PRNG sequences are periodic; they will eventually repeat themselves after generating a finite number of values.

Types of PRING:

- **Linear Congruential Generators (LCG):** Generates pseudo-random numbers using a simple, fast discontinuous piecewise linear equation based on a multiplier, an increment, and a modulus ($X_{n+1} = (aX_n + c)$ (mod m).
- **Mersenne Twister (MT19937):** Utilizes a matrix linear recurrence over a finite binary field to deliver a massive period of 2^{19937} - 1 and high dimensional uniformity, making it standard for general simulations but unsafe for cryptography.
- **Permuted Congruential Generator (PCG):** Improves upon the LCG architecture by applying a variable-dependent bitwise permutation function to the output,

destroying structural defects to achieve excellent statistical quality and high speed.

Cryptographic random number generation (CRNG):

- is a deterministic algorithm designed to produce sequences of numbers that are completely indistinguishable from true physical randomness by any adversary with bounded computing power.
- The Next-Bit Test (Forward Security): Given the first k bits of an output sequence, there is no polynomial-time algorithm that can predict the (k+1)-th bit with a probability greater than 50%.
- State Compromise Extension (Backward Security / Forward Secrecy): If an attacker manages to steal or deduce the internal state of the generator at a specific time t, it must be mathematically impossible for them to reconstruct the state or the output sequence generated prior to time t.

Types of CRNG:

- **Hash-Based Counter Generators:** Produces secure random bits by passing a combined seed and a continuously incrementing counter through a one-way cryptographic hash function like SHA-256.
- **Block Cipher-Based Generators:** Utilizes symmetric encryption algorithms (such as AES in CTR mode) to encrypt a steadily advancing counter, turning the resulting ciphertext into a random bitstream.
- **Stream Cipher-Based Generators:** Employs high-speed symmetric stream ciphers (like ChaCha20) to rapidly mutate an internal state array via bitwise operations and key rotations, producing an unpredictable output stream.
- **Number-Theoretic (Asymmetric) Generators:** Bases its security directly on hard, unsolved mathematical problems (like integer factorization in Blum Blum Shub), offering high theoretical security at the cost of slow computational speed.
- **HMAC-Based Generators:** Applies an ongoing feedback loop using Hash-based Message Authentication Codes to constantly mix and digest internal keys and state values, ensuring robust forward and backward security.

Feature / Dimension	True Random Number Generator (TRNG)	Pseudo-Random Number Generator (PRNG)	Cryptographic Random Number Generator (CRNG)
Core Mechanism	Captures unpredictable, chaotic, and non-deterministic physical processes.	Uses deterministic mathematical formulas to produce a sequence.	Uses a deterministic algorithm designed to mimic true physical randomness against bounded computing power.
Source of Randomness	A physical entropy source (e.g., quantum fluctuations, thermal noise, atmospheric static).	An initial input value called a seed.	A deterministic cryptographic structure (e.g., hashes, ciphers, or hard mathematical problems).
Predictability & Reproducibility	Cannot be predicted or reproduced, even with complete hardware knowledge.	Completely reproducible every single time if the seed and algorithm are known.	Completely unpredictable; passes the Next-Bit Test (cannot predict the next bit with >50% probability).
Periodicity (Repetition)	Non-deterministic and inherently chaotic (no mathematical period mentioned).	Periodic; the sequences will eventually repeat themselves after a finite number of values.	Deterministic, but sequences are completely indistinguishable from true physical randomness.
State Compromise Vulnerability	Unaffected by state compromise because it relies on spontaneous physical decay or chaotic environments.	Not inherently protected; if the internal algorithm/seed is known, the entire sequence is exposed.	Protected by State Compromise Extension; impossible to reconstruct past states/outputs even if the current internal state is stolen.
Security Level	Secure against attackers due to its chaotic physical nature.	Generally unsafe for cryptography (e.g., Mersenne Twister).	High security; specifically designed to withstand adversaries with bounded computing power through forward and backward security.
Computational Speed Characteristics	Bound by physical limits (e.g., speed of atomic degradation or hardware toggling).	Typically fast, utilizing simple or matrix linear equations.	Can vary; ranges from high-speed stream ciphers to slow computational speeds for number-theoretic generators.